

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2019

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2019 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Write 4000000 in standard form. [2]
2. Evaluate $8 + 3 \times 3 - 6$. [2]
3. In a class of Moleli, $n(A) = 17$ (play soccer), $n(A \cap B) = 5$ (play both). How many play soccer only? [3]
4. Find the HCF and LCM of 18 and 16. [4]
5. Express 18 as a percentage of 40. [3]
6. Expand and simplify $2(x + 1) - 5x$. [3]
7. A map of Mutare uses scale 2 : 800000. A road is 10 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm $\leftrightarrow$ 1 km if 2:8 is first simplified — instead: scale 2 cm to 8 km.) [4]
8. Overnight in Mutare the temperature was 0°C . By 14:00 it was 27°C . Find the rise. [3]

- 9.** A tank in Sakubva holds 40 litres. It is $\frac{2}{4}$ full. How many litres are in the tank? **[3]**
- 10.** Simplify $4^4 \times 4^3$. Leave as a power of 4. **[3]**
- 11.** Simple interest on \$200 at 10% per year for 2 years. Find the interest. **[4]**
- 12.** Circumference of a circle radius 14 cm. Take $\pi = \frac{22}{7}$. **[4]**
- 13.** Work out $(4 \ 3 ; 1 \ 0)(5 ; 2)$. **[4]**
- 14.** Factorise $x^2 + 4x + 3$. **[4]**
- 15.** Make h the subject of $A = 2\pi r(r + h)$. **[4]**
- 16.** Solve $\begin{cases} x + y = 11 \\ x - y = 1 \end{cases}$. **[4]**
- 17.** A cuboid water tank measures 12 m by 7 m by 7 m. Find its volume. **[3]**
- 18.** A bag contains 6 red, 5 blue and 4 green beads. $P(\text{not green})$? **[3]**
- 19.** A sequence is 8, 11, 14, 17, ... Find an expression for the n th term. **[4]**
- 20.** A regular polygon has 6 sides. Find each exterior angle. **[3]**

- 21.** In right-angled triangle XYZ, $\hat{X} = 60^\circ$ and hypotenuse $XZ = 12$ cm. Find XY, adjacent to 60° . **[4]**
- 22.** A triangular plot has base 6 m and perpendicular height 11 m. Find its area. **[3]**
- 23.** Find the mean of 10, 4, 9, 8, 4. **[3]**
- 24.** Solve $3(x + 1) = 12$. **[4]**
- 25.** Find the midpoint of (4, 6) and (3, 3). **[3]**
- 26.** Solve $4x + 1 < 25$. **[3]**
- 27.** A ladder of length 20 m leans against a wall. The foot is 12 m from the wall. How far up the wall does it reach? **[4]**
- 28.** Factorise completely $30x + 30$. **[3]**
- 29.** In triangle ABC, $\hat{A} = 47^\circ$ and $\hat{B} = 65^\circ$. Find $\hat{C}$. **[3]**
- 30.** Find the gradient of the line joining (4, 4) and (7, 10). **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).